BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 8465

Retyped faithfully from the original board paper — same questions, no changes.

Biology - Paper: II - Objective (iii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8465

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. The biotic components that obtain their energy from dead decaying materials are:

(A) Producers
(B) Primary consumers
(C) Secondary consumers
(D) Decomposers

Q2. In terrestrial vertebrates, gill pouches develop into _____.

(A) Gills
(B) Middle ear
(C) Eustachian tube
(D) Auditory tube

Q3. Organic chemical needed to make aspartame is:

(A) phenylalanine
(B) Alanine
(C) Proline
(D) Valine

Q4. Thermus aquaticus is a:

(A) RNA polymerase
(B) DNA ligase
(C) RNA primase
(D) DNA polymerase

Q5. Two normal parents have an albino child. What is the probability that their next child will also be an albino?

(A) 50 %
(B) 75 %
(C) 25 %
(D) 100 %

Q6. Persons affected from Turner's syndrome have _____ chromosomes.

(A) 44 Autosomes - X
(B) 44 Autosomes + X
(C) 44 Autosomes + Y
(D) 44 Autosomes - Y

Q7. Which are not present in RNA?

(A) Adenine and thymine
(B) Thymine and deoxyribose
(C) Uracil and deoxyribose
(D) Uracil and ribose

Q8. In prokaryotes, the promoter binding sequence TTGACA is also called _____ sequence.

(A) -35
(B) -10
(C) -75
(D) -25

Q9. Yellow cytoplasm of ascidian egg produces:

(A) Epidermis
(B) Gut
(C) Muscle cells
(D) Neural tube

Q10. The most effective temperature for vernalisation is:

(A) 5°C
(B) 10°C
(C) 7°C
(D) 4°C

Q11. Which of the given share a common hypothalamic releasing factor?

(A) LH and FSH
(B) LH and Prolactin
(C) TSH and FSH
(D) FSH and Prolactin

Q12. Which are the growth promoter hormones?

(A) Cytokinins and abscisic acid
(B) Auxins and ethene
(C) Auxins and cytokinins
(D) Auxins and abscisic acid

Q13. What happens during muscle contraction?

(A) I-band shortens, H-zone disappears
(B) I-band shortens, Z-lines brought away
(C) Z-lines brought closer, H-zone maximizes
(D) I-band elongates, H-zone disappears

Q14. Which bones are part of pelvic girdle?

(A) Ilium and scapula
(B) Ischium and pubis
(C) Pubis and suprascapula
(D) Ischium and clavicle

Q15. How much water is needed to eliminate 1g of ammonia nitrogen?

(A) 1 ml
(B) 50 ml
(C) 100 ml
(D) 500 ml

Q16. Which component of CFCs destroys ozone?

(A) Carbon
(B) Fluorine
(C) Chlorine
(D) All of these

Q17. Which biome has very fertile and rich organic matter soil?

(A) Deciduous forests
(B) Grass land
(C) Deserts
(D) Tundra

Biology - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note:- Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. What is a flame cell?

ii. Define heat shock proteins.

iii. How do terrestrial animals dispose of excess heat from the body? Discuss it.

iv. How does a plant cell maintain turgor pressure by tonoplast? Comment on it.

v. How does secondary xylem produce growth rings during secondary growth?

vi. How is callus formed on damaged surfaces of stem and root? Give its significance.

vii. Ovulation in human female is due to hormonal control. Comment on it.

viii. What are fraternal twins?

ix. What is Eutrophication?

x. Distinguish Littoral zone from Limnetic zone.

xi. What is a nutrient cycle?

xii. What are industrial effluents? Give their any two effects.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Write two physiological roles of gibberellins.

ii. What is Parkinson's disease? Write its treatment.

iii. Define effectors. Give examples.

iv. Compare protanopia with deuteranopia.

v. Differentiate between homozygote and heterozygote.

vi. Justify, why AB blood group is a universal recipient?

vii. What is a probe? Give its importance.

viii. What is Taq polymerase? Give its significance.

ix. Why urine is preferable than milk for biotechnology?

x. Differentiate between primary and secondary succession.

xi. Define biome and biosphere.

xii. How nitrogen is depleted and what are its remedies?

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. How is coelom formed in chick embryo?

ii. Compare teratology with gerontology.

iii. What is phosphodiester bond? Sketch it.

iv. Differentiate between sense strand and antisense strand of DNA.

v. What is phenylketonuria? What kind of disorder is it?

vi. What is equatorial plate? When is it formed in meiosis?

vii. How does cytokinesis occur in animal cells?

viii. What is endosymbiotic hypothesis? Give an example.

ix. How does genetic drift affect the gene frequency?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT MARKS) (4+4=8) (8 x 3 = 24)

Q5.. (Marks 8) (a) Discuss kidney stone formation and its treatment. (b) Write a note on cancer (Uncontrolled cell division).

Q6.. (Marks 8) (a) Describe the structure of axial skeleton. (b) Explain the components of ecosystem.

Q7.. (Marks 8) (a) How is nervous system of planaria better developed than that of Hydra? (b) How does comparative anatomy support evolution?

Q8.. (Marks 8) (a) How sexual reproduction is different from asexual reproduction? Give a detailed answer. (b) State Mendel's law of segregation. Explain this law with the help of an example.

Q9.. (Marks 8) (a) What is aging? Describe its causes and symptoms. (b) What are restriction endonucleases? Give their role in recombinant DNA technology.